Higher National Diploma in Engineering (Electrical)
Fourth Year, First Semester Examination – 2023

EE 3212– Broadcast Technologies

Instructions for Candidates:	No. of Questions: **06**
Answer **FIVE (05)** questions	No. of Pages : **03**
All questions carry equal marks	Time : **(03) Three Hours**

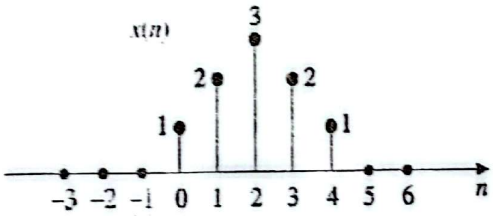
Instruction to Candidates: Answer **Q1** and any other four questions. **Q1 is compulsory.**

01.

i. Briefly describe the terms with examples '*signals*' and '*systems*' in *Digital Signal* Processing. **(04 Marks)**

ii. What are the difference between *discrete-time unit step sequence u(n)* and *the shifted version of the discrete-time unit step sequence u (n – k)*? Explain with graphical representation. **(04 Marks)**

iii. An arbitrary signal x(n) is shown in Figure below.



Show the time-shifting graphic representation following below. **(03 Marks)**

a) x(n-3)
b) x(n+2)

iv. Sketch the following signals.

a) x(n) = u(n+3) u(-n+2)
b) x(n) = u(n+4) – u(n-2) **(06 Marks)**

v. Consider a single x (n) with values as follows,

x (-2) = -3, x (-1) = 2, x (0) = 0, x (1) = 3, x (2) = 1 and x (3) = 2

Deliver the above-mentioned discrete-time signal ***a functional and graphical*** representation. **(03 Marks)**

[Total 20 Marks]

02.

i. Briefly explain the term of *Television Broadcasting* clearly. **(03 Marks)**

ii. How does differentiate *Analog TV* and *Digital TV*. **(04 Marks)**

iii. Briefly describe the following recoding method using suitable diagrams. **(2x3=06 Marks)**

a. X-Y technique
b. A-B technique
c. M/S technique

iii. How is stereo sound recorded? Explain. **(03 Marks)**

iv. How matrixing work in FM stereo broadcasting? Clearly explain. **(04 Marks)**

[Total 20 Marks]

03.

i. Compare between *DSB-SC* (Double Side Band Suppressed Carrier Modulation) and *AM* (Amplitude Modulation)? **(04 Marks)**

ii. Why is *AM* commonly used for radio broadcasting instead of *DSB-SC*? **(04 Marks)**

iii. Define *Pre-Emphasis* and *De-Emphasis* in *VHF* broadcast. **(04 Marks)**

iv. Briefly explain *Cross talk between L and R channels*. **(04 Marks)**

v. What is meant by *RDS* in FM radio broadcasts? And what are data including? **(04 Marks)**

[Total 20 Marks]

04.

i. Define the terms ***scanning*** and ***scanning process*** in television clearly. **(04 Marks)**

ii. Explain the terms following with suitable sketches. **(08 Marks)**

a) Horizontal Scanning
b) Vertical Scanning
c) Progressive Scanning
d) Interlaced Scanning.

iii. What is the role of the electron beam in scanning for cathode-ray tube (CRT) televisions? **(04 Marks)**

iv. What is meant by *Aspect Ratio* and describe reasons for having this ratio? **(04 Marks)**

[Total 20 Marks]

05.

i. Describe the terms *hue, saturation* and *flicker* with examples. **(06 marks)**

ii. What is the role of photoconductivity in TV? **(03 marks)**

iii. What is the main function of the blanking pulses? **(03 Marks)**

iv. Define picture resolution. **(02 Marks)**

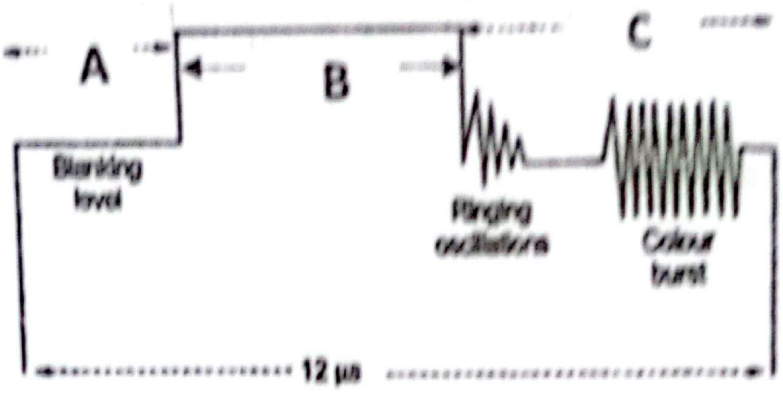
v. How does obtained *complementary colors* in TV? **(03 Marks)**

vi. Differentiate *additive color mixing* and *subtracting color mixing*. **(03 Marks)**

[Total 20 Marks]

06.

i. The following figure shows the '**Horizontal Blanking Pulse**'. Name A, B and C with their time periods. **(03 Marks)**



ii. Why back porch is longer than front porch? Explain. **(03 Marks)**

iii. State the *'Grassman's law'*. **(02 Marks)**

iv. Give the applications of *Video Tape Recorders* (VTR). **(03 Marks)**

v. What are the drawbacks of *Delta Gun Picture Tube*? **(03 Marks)**

vi. Explain how a monochrome TV transmitter works with a suitable diagram. **(06 Marks)**

[Total 20 Marks]